

Structured Retrieval for Verifiable Scientific Question Answering

Abstract

This paper presents a pipeline for grounded question answering over scientific literature. We combine a heuristic section tree with a BM25 retrieval layer over a knowledge graph of extracted claims, and show that the resulting citations remain verifiable against source text.

1. Introduction

Scientific papers are long and unevenly structured, which makes locating the right evidence for a claim difficult for both humans and automated systems. We study how a lightweight, fully offline pipeline can turn a raw PDF into retrievable, provenance-tracked units.

2. Methods

Our system builds a knowledge graph from parsed sections and ranks candidate passages with BM25, a classical sparse ranking method that requires no network access or embedding model. Each retrieval unit records its exact character offsets so every ranked result can be traced back to the original document text.

3. Results

We evaluate the pipeline on a multi-hop question answering benchmark we call SciQA-Mini, a small curated set of research questions that each require combining evidence from two or more sections. The sparse retrieval layer alone recovers the correct section for the large majority of SciQA-Mini questions without any learned ranker.

4. Conclusion

Grounding answers in verifiable citations keeps the system honest: every claim can be traced to a specific passage in the source PDF. We argue that this offline-first design is a practical baseline for research assistants that must work without external services.

References

- [1] A. Researcher and B. Scholar. Sparse and Structured Retrieval for Scientific Papers. *Journal of Made-Up Studies*, 2026.